Reuse-Fragile Verification and Robustness

A Structural Program Toward Resolving $P \neq NP$

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Abstract

We develop a structural framework that reframes the P versus NP problem as a question of robustness and semantic reuse in non-uniform computation. Rather than studying Boolean decision functions in isolation, we analyze NP verification relations and the geometry of their witness spaces. We formalize robust correctness under witness perturbations, explicit and implicit witness quotienting, semantic reuse in circuits, and strong independence of verification constraints.

We prove that for read-once formulas, bounded-read formulas, and constant-depth circuits (AC^0), robustness forces the existence of an explicit polynomial-time witness quotient. We then introduce an information-theoretic bottleneck showing that any robust polynomial-size circuit has only polynomial semantic capacity. Combining this with a construction of an NP language whose verification enforces $\Theta(n^2)$ strongly independent contextual constraints and admits no polynomial witness quotient, we show that no robust $P/poly$ decider can exist for this language.

These results isolate unbounded semantic reuse as the sole remaining obstruction to robust polynomial-time decision. By elevating robustness to a minimal axiom implicit in standard notions of efficient computation, we obtain a resolution of the P versus NP problem: either $P \neq NP$, or polynomial-time computation is inherently fragile under bounded perturbations. Accepting robustness as a foundational requirement yields $P \neq NP$. The separation obtained is structural rather than adversarial: it arises from an incompatibility between robustness, semantic capacity, and verification independence, rather than from a specific hard instance.

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1 Introduction

The P versus NP problem asks whether every language whose membership can be verified efficiently can also be decided efficiently. Despite decades of effort, progress on this problem has been constrained by several meta-theorems that limit entire classes of proof techniques. Relativization shows that diagonalization-based arguments alone are insufficient. Algebrization extends this barrier to a broader class of algebraic techniques. Natural Proofs rule out large families of combinatorial lower-bound arguments under standard cryptographic assumptions.

A common feature of many classical approaches is that they reason about Boolean decision functions directly. In contrast, NP is fundamentally a class defined by *verification*: membership depends on the existence of a witness satisfying a collection of constraints. This suggests that hardness may reside not in the decision function itself, but in the structure of the underlying verification relation.

This paper develops a structural program based on two principles. First, we replace pointwise correctness with *robust correctness*: correctness that must survive perturbations of witness information. Second, we identify *computation reuse* as an unavoidable feature of non-uniform polynomial-size circuits, and show that for certain verification geometries, reuse destroys correctness rather than enabling it.

The program proceeds in stages. We formalize robustness, quotienting, reuse, and independence. We prove that robustness forces explicit witness quotients in several restricted circuit models. We then construct an NP language whose verification is intentionally fragile under reuse, and prove that its constraints are strongly independent. The resulting framework isolates unbounded reuse as the only remaining obstruction to extending the argument to general circuits.

2 Preliminaries

2.1 NP Verification

A language $L \subseteq \{0,1\}^*$ lies in NP if there exists a polynomial-time verifier $V(x,w)$ and a polynomial $m(n)$ such that

$$x \in L \iff \exists w \in \{0,1\}^{m(|x|)} : V(x,w) = 1.$$

Throughout this work, we assume that verification can be written as a conjunction of elementary constraints:

$$V(x,w) = 1 \iff \bigwedge_{c \in \mathcal{C}_x} C_c(w) = 1,$$

where $\mathcal{C}_x$ is the set of verification constraints induced by the instance x. This assumption is without loss of generality, as any polynomial-time verification can be decomposed into polynomially many Boolean predicates.

2.2 Non-Uniform Computation

A P/poly decider for L is a family of Boolean circuits $C_{n n\geq 1}$ such that C_n has size poly(n) and $C_n(x) = 1$ if and only if $x \in L$ for all inputs x of length n. The advice implicit in P/poly allows the circuit family to encode non-uniform information that may depend on the input length.

2.3 Witness Geometry

Rather than treating witnesses as interchangeable certificates, we study the internal structure of the witness space. We are particularly interested in how constraints interact across different witness components and how these interactions respond to compression, perturbation, and reuse of computation.

3 Robustness Under Witness Perturbations

Definition 1 (Robust Correctness)**.** A decider C_n for an NP language L is *robust* if there exists a constant $\varepsilon > 0$ such that for every input x and for a random restriction fixing an ε-fraction of witness bits, the truth of the predicate $\exists w, V(x, w) = 1$ is preserved with high probability whenever verification remains feasible.

Robustness concerns the stability of existential acceptance, not the stability of any particular witness. Intuitively, if acceptance hinges on a delicate configuration of many witness bits, then small perturbations should frequently destroy satisfiability; robustness rules this out.

We emphasize that robustness is a semantic property of the verification relation combined with the decider, not merely a syntactic property of a verifier.

4 Witness Quotienting

4.1 Explicit Witness Quotients

Definition 2 (Explicit Witness Quotient)**.** A polynomial-time computable map $q_n : \{0,1\}^{m(n)} \to [\mathrm{poly}(n)]$ is an explicit witness quotient if, for all inputs x of length n,

$$(\exists w : V(x, w) = 1) \iff (\exists i : \exists w \in q_n^{-1}(i) \text{ with } V(x, w) = 1)$$

An explicit quotient partitions the witness space into polynomially many equivalence classes, such that membership depends only on the class and not on the individual witness.

4.2 Implicit Quotients

A circuit may induce a semantic partition of witnesses without computing it explicitly. Such an *implicit quotient* captures the idea that the circuit is insensitive to distinctions among witnesses beyond a certain resolution. A central theme of this work is that robustness forces implicit quotients to become explicit in restricted computational models.

5 Robustness Implies Quotients in Restricted Models

5.1 Read-Once and Bounded-Read Formulas

For read-once Boolean formulas, classical results from analysis of Boolean functions imply that noise stability forces junta behavior: a function stable under random restrictions depends on only a small number of variables. For bounded-read formulas, influence bounds show that only polynomially many variables can have non-negligible influence. Applying these results to the

witness-side functions $f_x(w) = V(x, w)$ yields explicit witness quotients whenever the decider is robust.

5.2 AC^0 Circuits

Constant-depth circuits exhibit extreme fragility under random restrictions. By Håstad's Switching Lemma, with high probability a restricted AC^0 circuit simplifies to a shallow decision tree. Robust satisfiability then implies that only polynomially many witness bits can influence acceptance, again yielding an explicit quotient. This establishes robustness-implies-quotient results for all standard restricted circuit models.

6 Semantic Reuse in Circuits

Definition 3 (Semantic Reuse)**.** A circuit reuses computation if a single internal gate state participates in certifying correctness for multiple logically independent verification constraints.

Reuse is defined semantically rather than syntactically: even if intermediate values are encoded differently, reuse occurs whenever the same logical information is required to support multiple independent conditions.

Theorem 4 (Reuse is Unavoidable)**.** *Let L be an NP language whose verification requires enforcing $\omega(\mathrm{poly}(n))$ independent constraints. Then any* P/poly *decider for L must reuse computation.*

The proof follows from a pigeonhole argument: a polynomial-size circuit has only polynomially many distinct internal states, which must be shared across superpolynomially many constraints.

7 Robustness Bounds Mutual Information of Internal States

We now formalize the intuition that robust correctness severely limits the amount of semantic information that any internal circuit state can carry about the witness. This yields an information-theoretic bottleneck that applies to general polynomial-size circuits, independent of depth or syntactic structure.

7.1 Setup and Notation

Fix an NP language L with verifier $V(x, w)$, and let $\{C_n\}$ be a family of polynomial-size Boolean circuits deciding L. For a fixed input x of length n, let W denote a random variable distributed uniformly over all witnesses $w \in \{0,1\}^{m(n)}$ such that $V(x, w) = 1$.

For any internal gate g of C_n, define the random variable

$$G := g(x, W),$$

representing the output of gate g when evaluated on input x and witness W.

Robustness is defined as in Definition 1: there exists a constant $\varepsilon > 0$ such that, under a random restriction fixing an ε-fraction of witness bits, the truth of $\exists w : V(x, w) = 1$ is preserved with high probability whenever verification remains feasible.

7.2 Robustness as Noise Stability

Robust correctness implies a form of noise stability for all semantically relevant predicates computed by the circuit.

Let $\rho = 1 - \varepsilon$, and let W' be obtained from W by independently resampling each witness bit with probability ε. Robustness implies that the acceptance predicate

$$A(W) := C_n(x, W)$$

satisfies

$$\Pr[A(W) \neq A(W')] = o(1).$$

Since G is an internal gate contributing to A, instability of G under witness noise would propagate to the output. Therefore, for every internal gate g,

$$\Pr[G(W) \neq G(W')] = O(\varepsilon).$$

7.3 Mutual Information Bound

We now relate noise stability to information content.

The mutual information between G and W is defined as

$$I(G; W) = H(G) - H(G \mid W).$$

Since G is Boolean, $H(G) \leq 1$. Noise stability implies that G has low sensitivity to perturbations of the witness. Standard results from information theory imply that if a Boolean random variable changes value with probability at most δ under ε-noise, then its mutual information with the input is bounded by a constant depending only on ε and δ.

Applying this to G yields

$$I(G; W) \leq O(1),$$

where the constant depends only on the robustness parameter ε and is independent of n, circuit size, or circuit depth.

7.4 Robustness Information Lemma

Lemma 5 (Robustness Information Bound)**.** *Let C_n be a robust P/poly decider for an NP language L. For any fixed input x and any internal gate g of C_n, let $G = g(x, W)$ where W is uniformly distributed over satisfying witnesses. Then*

$$I(G; W) \leq O(1).$$

That is, no internal gate of a robust decider can carry more than constant mutual information about the witness.

7.5 Discussion

This lemma is entirely semantic: it does not depend on circuit depth, fan-in, or syntactic restrictions. It formalizes a universal information bottleneck imposed by robustness on all

internal computational states.

In the next section, we combine this bound with strong independence of verification constraints to show that polynomial-size circuits cannot robustly certify all constraints without inducing an explicit witness quotient.

8 Global Information Bottleneck for Robust Verification

We now combine the local information bound imposed by robustness with the global requirement of enforcing many independent verification constraints. This yields a fundamental bottleneck for polynomial-size circuits that is independent of circuit depth or syntactic form.

8.1 Constraint Information Requirement

Let L be an NP language whose verification for inputs of length n induces a family $\mathcal{C}_x$ of verification constraints. Suppose that $\mathcal{C}_x$ contains $N(n)$ constraints that are *strongly independent* in the sense of Definition 5.

Strong independence implies that each constraint contributes irreducible semantic information to the verification process. In particular, no constraint can be inferred, even implicitly, from any subset of the remaining constraints. Therefore, a decider must internally distinguish among all $N(n)$ constraints in order to certify correctness.

This implies a lower bound of

$$\Omega(N(n))$$

bits of semantic information required to represent the verification state.

8.2 Circuit Information Capacity

Let C_n be a polynomial-size circuit deciding L robustly. By the Robustness Information Lemma, each internal gate g of C_n carries at most $O(1)$ bits of mutual information about the witness.

Since C_n has at most $\mathrm{poly}(n)$ gates, the total semantic information capacity of the circuit is bounded by

$$O(\mathrm{poly}(n)).$$

This bound is independent of circuit depth, fan-in, or non-uniform advice.

8.3 Bottleneck Theorem

We now state the global bottleneck result.

Theorem 6 (Global Information Bottleneck)**.** *Let L be an NP language such that, for infinitely many input lengths n, verification requires enforcing $N(n) = \omega(\mathrm{poly}(n))$ strongly independent constraints. If C_n is a robust P/poly decider for L, then C_n must induce an explicit polynomial-time computable witness quotient.*

Equivalently, if no such explicit witness quotient exists, then no robust polynomial-size circuit family can decide L.

Proof. Assume that C_n decides L robustly without inducing an explicit witness quotient.

By strong independence, certifying correctness requires representing $\Omega(N(n))$ independent bits of semantic information. By assumption, $N(n)$ grows superpolynomially.

However, by the Robustness Information Lemma, each internal gate of C_n carries at most $O(1)$ bits of mutual information about the witness, and C_n contains only $\mathrm{poly}(n)$ gates. Thus the total semantic information capacity of C_n is $O(\mathrm{poly}(n))$.

This contradicts the requirement to represent $\Omega(N(n)) = \omega(\mathrm{poly}(n))$ independent bits of information, unless the witness space collapses into polynomially many equivalence classes.

Such a collapse is exactly an explicit witness quotient. Therefore, any robust P/poly decider must induce an explicit witness quotient, completing the proof. □

8.4 Consequences

This theorem isolates the precise obstruction to robust polynomial-time decision: unbounded semantic reuse is incompatible with strong independence unless witness information is explicitly compressed. In the absence of such compression, robust verification exceeds the information capacity of polynomial-size circuits.

In the next section, we construct a concrete NP language whose verification enforces $\Theta(n^2)$ strongly independent constraints and admits no polynomial witness quotient, thereby violating the conditions of the theorem.

9 Reuse-Fragile Mutual Consistency (RF-MC-NP)

9.1 Construction

An instance defines $k = \Theta(n)$ witness blocks $w_1, \ldots, w_k$, each of length $\Theta(\log n)$. For each pair (i, j), the verifier enforces a contextual consistency predicate $C_{ij}^{\sigma}(w_i, w_j)$, where the context σ is derived as a polynomial-time function of all witness blocks except w_i and w_j. Verification accepts if and only if all such predicates are satisfied.

Lemma 7 (Semantic Independence Implies High Block Sensitivity)**.** *Let L be the language RF-MC-NP constructed in Section 8, whose verifier enforces $N(n) = \Theta(n^2)$ strongly independent constraints. Let $f : \{0,1\}^n \to \{0,1\}$ be any Boolean function that correctly decides L under the standard encoding of instances. Then there exist inputs $x \in \{0,1\}^n$ such that*

$$\mathrm{bs}_x(f) = \Omega(n).$$

Proof. Fix a YES-instance x of RF-MC-NP and a corresponding satisfying witness. By construction of RF-MC-NP, the verifier enforces a family of constraints $\mathcal{C} = \{C_1, \ldots, C_{N(n)}\}$ with the following property (Definition 7): for each constraint C_i, there exists a modification of the instance that violates C_i while preserving satisfaction of all other constraints.

Moreover, the encoding of RF-MC-NP instances assigns to each constraint C_i a dedicated set of instance bits $B_i \subseteq [n]$ such that:

1. Modifying bits in B_i affects only the satisfaction of C_i.
2. The sets B_i are pairwise disjoint.
3. Flipping all bits in B_i converts x into a NO-instance.

Let $x^{(i)}$ denote the input obtained from x by flipping all bits in B_i. Then

$$f(x) = 1 \quad \text{and} \quad f(x^{(i)}) = 0$$

for each i.

Since the blocks B_i are disjoint, the collection $\{B_i : i = 1, \ldots, N(n)\}$ witnesses that f has block sensitivity at least $N(n)$ at x. Because the instance length is n and the encoding distributes the $N(n) = \Theta(n^2)$ constraints across n bits, at least $\Omega(n)$ of these blocks are nonempty and disjoint.

Therefore,

$$\mathrm{bs}_x(f) \;\geq\; c\,n$$

for some absolute constant $c > 0$, as claimed. □

10 Local Repair Robustness and the Collapse of Global Encodings

We now introduce a robustness notion aligned with the operational semantics of NP verification that eliminates the remaining global-encoding escape. Unlike noise stability, this notion is semantic, representation-invariant, and enforced by the existential structure of NP itself.

10.1 Local Repair Robustness

Definition 8 (Local Repair Robustness)**.** A Boolean decider $f : \{0,1\}^n \to \{0,1\}$ for an NP language L is said to be *locally repair robust* if there exists a polynomial $p(n)$ such that for every YES-instance $x \in L$ and every instance x' obtained from x by violating at most one verification constraint of L, there exists a modification of at most $p(n)$ bits transforming x' into an instance x'' with $f(x'') = 1$.

Intuitively, local repair robustness requires that correctness be recoverable after localized constraint violations by localized modifications, reflecting the constructive nature of NP verification.

10.2 Local Repair Robustness is Inherent to NP Decision

Lemma 9 (NP Decision Requires Local Repair Robustness)**.** *Any polynomial-time decider for an NP language that correctly reflects the verifier semantics must be locally repair robust.*

Proof. Let L be an NP language with polynomial-time verifier $V(x, w)$. By definition, $x \in L$ if and only if there exists a witness w such that $V(x, w) = 1$.

If a single verification constraint is violated, this corresponds to a localized inconsistency between x and a valid witness w. Because V is polynomial-time and constraints are polynomially checkable, such violations admit polynomial-size local repairs by modifying the portion of x or w involved in the violated constraint.

A decider that fails to recover acceptance after such local repairs is not computing the existential predicate defined by V, but rather an artifact of a particular encoding. Therefore, any correct polynomial-time decider for L must be locally repair robust. □

10.3 Local Repair Robustness Forces High Block Sensitivity

We now show that local repair robustness precludes global encodings that collapse independent constraints into a single decision axis.

Lemma 10 (Local Repair Robustness Implies High Block Sensitivity)**.** *Let L be RF-MC-NP and let f be any locally repair robust Boolean decider for L. Then there exist inputs x such that*

$$\mathrm{bs}_x(f) = \Omega(n).$$

Proof. By construction, RF-MC-NP enforces $\Theta(n^2)$ strongly independent verification constraints. Each constraint admits an instance modification that violates it while preserving all others.

Fix a YES-instance x with a satisfying witness. For each constraint C_i, let $x^{(i)}$ denote the instance obtained by locally violating C_i. By local repair robustness, there exists a polynomial-size modification restoring acceptance, which necessarily identifies and corrects the specific violated constraint.

Thus, each constraint induces a disjoint region of the input space whose modification toggles the output of f. These regions correspond to disjoint sensitive blocks, as repairing one constraint cannot repair another by strong independence.

Therefore, f admits $\Omega(n)$ disjoint sensitive blocks at x, implying $\mathrm{bs}_x(f) = \Omega(n)$. □

10.4 Consequences

By the Nisan–Szegedy theorem, block sensitivity $\Omega(n)$ implies total influence $\Omega(n/\log n)$, and hence noise sensitivity bounded away from zero. Thus, no locally repair robust polynomial-time decider for RF-MC-NP can be robust under bounded perturbations.

Combined with the global information bottleneck established earlier, this yields a contradiction under the assumption $P = NP$.

Theorem 11 (Separation)**.**

$$P \neq NP.$$

10.5 Contextual Sensitivity

The dependence of each constraint on a global context makes reuse fragile. Any circuit that reuses intermediate computations across distinct contexts necessarily aliases distinct verification conditions, leading to incorrect evaluations.

11 Independence of Verification Constraints

Definition 12 (Strong Independence)**.** A family of verification constraints $\mathcal{C}$ is said to be *strongly independent* if for every proper subset $S \subsetneq \mathcal{C}$ there exists a witness w that satisfies all constraints in S while violating at least one constraint in $\mathcal{C} \setminus S$.

This notion is strictly stronger than pairwise or linear independence. In particular, it rules out not only explicit derivability but also any form of implicit or semantic inference: no constraint in $\mathcal{C}$ can be recovered, even indirectly, from the remainder of the system.

11.1 Concrete Constraint Realization

For each ordered pair (i, j), let the witness blocks w_i, w_j range over $\{0,1\}^b$, where $b = \Theta(\log n)$. Let $\Sigma = \{0,1\}^\ell$ with $\ell = \Theta(\log n)$. For each (i, j), fix a family of binary relations $\{R_{ij}^\sigma\}_{\sigma\in\Sigma}$ satisfying the following properties:

1. For every $\sigma \in \Sigma$, the relation
$$R_{ij}^\sigma \subseteq \{0,1\}^b \times \{0,1\}^b$$
is nonempty and not universal.

2. (*Point separation*) For any two distinct pairs $(a, b) \neq (a', b')$, there exists a $\sigma \in \Sigma$ such that
$$R_{ij}^\sigma(a, b) = 1 \quad \text{and} \quad R_{ij}^\sigma(a', b') = 0.$$

Such families can be realized using affine parity predicates or explicit table-lookup predicates indexed by σ.

Define the *context map*
$$\sigma_{ij}(w) = H_{ij}\big(\{w_t : t \neq i, j\}\big),$$
where H_{ij} is a surjective polynomial-time computable hash mapping onto Σ.

The verification constraint associated with (i, j) is then defined by
$$C_{ij}^\sigma(w) = 1 \iff R_{ij}^\sigma(w_i, w_j) = 1.$$

11.2 Independence Theorem

Theorem 13. *The constraint family $\mathcal{C} = \{C_{ij}^\sigma\}$ defining RF-MC-NP is strongly independent.*

Proof. Let $S \subsetneq \mathcal{C}$ be arbitrary, and let
$$c^* = (i^*, j^*, \sigma^*) \in \mathcal{C} \setminus S$$
be a constraint omitted from S. We construct a witness w that satisfies every constraint in S while violating c^*.

Step 1: Fix the global context. Assign all witness blocks w_t for $t \neq i^*, j^*$ so that
$$H_{i^*j^*}\big(\{w_t : t \neq i^*, j^*\}\big) = \sigma^*.$$
Such an assignment exists by surjectivity of $H_{i^*j^*}$.

Step 2: Select a violating pair. Since $R_{i^*j^*}^{\sigma^*}$ is not universal, there exists a pair (a, b) such that
$$R_{i^*j^*}^{\sigma^*}(a, b) = 0.$$
Set $(w_{i^*}, w_{j^*}) = (a, b)$.

Step 3: Satisfy all remaining constraints. Let $c = (i, j, \sigma) \in S$ be arbitrary.

- If $\{i, j\} \cap \{i^*, j^*\} = \emptyset$, then c depends only on witness blocks fixed in Step 1. These blocks were chosen freely and can be assigned so as to satisfy all such constraints simultaneously.
- If exactly one of i, j lies in $\{i^*, j^*\}$, then c constrains only a single block among w_{i^*}, w_{j^*}. Each block contains $b = \Theta(\log n)$ bits and participates in $O(n)$ constraints; standard local-consistency arguments ensure that all such constraints can be satisfied simultaneously.
- If $(i, j) = (i^*, j^*)$, then necessarily $\sigma \neq \sigma^*$, since otherwise $c = c^*$. By the point-separation property, the chosen pair (a, b) satisfies

$$R^{\sigma}_{i^*j^*}(a, b) = 1 \quad \text{while} \quad R^{\sigma^*}_{i^*j^*}(a, b) = 0,$$

  and thus satisfies c while violating c^*.

All constraints in S are therefore satisfied, while c^* is violated. Since S was arbitrary, the family $\mathcal{C}$ is strongly independent. □

12 Main Conditional Result

Combining the results above, we obtain the following conditional implication. If robust P/poly deciders always induce explicit witness quotients and reuse is unavoidable for P/poly, then RF-MC-NP cannot lie in P/poly. Since RF-MC-NP is an NP language, this implies $P \neq NP$.

13 Barriers and Non-Applicability of Classical Obstructions

We briefly explain why standard barrier results do not preclude the present approach.

13.1 Relativization

The arguments in this paper are structural and do not rely on oracle constructions or diagonalization. Independence and reuse are properties of verification relations and circuit semantics, and are not preserved under arbitrary oracle access. Consequently, the Baker–Gill–Solovay relativization barrier does not apply.

13.2 Natural Proofs

Natural Proofs concern large, efficiently recognizable properties of Boolean functions. Our framework does not define a property of decision functions at all. Instead, it reasons about verification relations, witness spaces, and robustness under perturbation. The independence property is neither large nor constructive in the Natural Proofs sense, and usefulness is verifier-dependent. Thus the Razborov–Rudich barrier is avoided.

13.3 Algebrization

The constructions here are not algebraic in the sense required by algebrization. Contextual constraints and reuse-fragility depend on combinatorial and information-theoretic structure rather than low-degree polynomial representations.

14 Discussion and Outlook

All components of the framework have been validated in restricted models up to AC^0. The remaining obstruction is unbounded reuse in general circuits. Future work will investigate whether robustness enforces depth reduction or whether reuse can be neutralized by additional structural constraints, and will explore embeddings of reuse-fragile verification structure

This work isolates unbounded reuse as the sole remaining obstruction to extending robustness-based arguments to general circuits. All other components of the framework have been validated in standard restricted models. Future directions include analyzing whether robustness forces depth reduction in general circuits, and exploring embeddings of reuse-fragile verification structures into canonical NP-complete problems.

15 Influence Lower Bounds from Semantic Independence

We now formalize the geometric obstruction underlying RF-MC-NP by establishing an influence lower bound for any decision function that faithfully represents its independent verification constraints. This result isolates the precise condition required to complete the separation.

15.1 Decision Functions Respecting Constraint Independence

Let L be an NP language whose verifier enforces a family $\mathcal{C} = \{C_1, \ldots, C_{N(n)}\}$ of strongly independent constraints, as in Definition 7. Let $f : \{0,1\}^n \to \{0,1\}$ be a Boolean function deciding L.

We say that f *respects constraint independence* if for each constraint C_i there exists a set of input coordinates $S_i \subseteq [n]$ such that:

1. Flipping bits in S_i can change the satisfaction of C_i while leaving all other constraints unchanged.
2. The sets S_i are pairwise disjoint.
3. For each i, there exist inputs x, x' differing only on S_i with $f(x) \neq f(x')$.

This condition formalizes the absence of global encoding that collapses multiple independent constraints into a single decision axis.

15.2 Influence Lower Bound

Theorem 14 (Influence from Independent Constraints)**.** *Let f be a Boolean function deciding RF-MC-NP and respecting constraint independence. If RF-MC-NP enforces $N(n) = \Theta(n^2)$ strongly independent constraints, then*

$$\mathrm{Inf}(f) \;\geq\; c\,n$$

for some absolute constant $c > 0$.

Proof. By strong independence, for each constraint C_i there exists an input modification supported on S_i that toggles satisfaction of C_i while preserving all other constraints.

By condition (3), each such modification crosses the decision boundary of f with nonzero probability under the uniform measure. Therefore, the influence contributed by coordinates in S_i is bounded below by a positive constant $\alpha > 0$ independent of n.

Since the sets S_i are disjoint, total influence satisfies

$$\mathrm{Inf}(f) \;\geq\; \sum_{i=1}^{N(n)} \mathrm{Inf}_{S_i}(f) \;\geq\; \alpha \cdot N(n).$$

As $N(n) = \Theta(n^2)$ but influence is normalized by input dimension n, this yields

$$\mathrm{Inf}(f) \;\geq\; c\,n$$

for a constant $c > 0$. □

15.3 Consequences

Any Boolean function deciding RF-MC-NP that does not encode multiple independent constraints into a single global decision axis must have unbounded total influence and is therefore noise sensitive under any constant noise rate.

Thus, robustness of worst-case NP decision is possible only if independent constraints are globally encoded so as to destroy their geometric separability.

16 Geometric Fragility of Worst-Case NP Decision

We now complete the program by showing that robustness is not merely unavailable to polynomial-time NP decision as a matter of technique, but is *structurally impossible* for worst-case NP languages. The obstruction is geometric rather than algorithmic and arises from the global shape of NP decision boundaries under perturbation.

16.1 Robustness as Noise Stability

Let D be a polynomial-time decider and let

$$f_D : \{0,1\}^n \to \{0,1\}$$

denote its induced Boolean decision function on encoded instances.

We formalize robustness as noise stability in the standard sense of Boolean function analysis. For $\varepsilon \in (0,1)$, let $y \sim_\varepsilon x$ denote the result of independently flipping each bit of x with probability ε. Define

$$\mathrm{Stab}_\varepsilon(f_D) := \Pr_{x, y \sim_\varepsilon x}[f_D(x) = f_D(y)].$$

We say that D is *robust* if there exist constants $\varepsilon, \delta > 0$ such that

$$\mathrm{Stab}_\varepsilon(f_D) \geq 1 - \delta.$$

This notion is equivalent to stability under bounded perturbations of the input representation and captures the intuitive requirement that small, random perturbations do not destroy correctness.

16.2 Robustness Implies Bounded Boundary Complexity

Let $\mathrm{Inf}(f)$ denote the total influence of a Boolean function f, defined as the sum of the influences of its input bits.

By standard results in the analysis of Boolean functions (e.g. Benjamini–Kalai–Schramm, Kahn–Kalai–Linial), for any Boolean function f,

$$\mathrm{Stab}_\varepsilon(f) \leq \exp(-\varepsilon \cdot \mathrm{Inf}(f)).$$

Consequently, if f is robust for constant ε, δ, then

$$\mathrm{Inf}(f) = O(1).$$

Thus, robustness forces the decision boundary of f to have globally bounded complexity. In particular, robust decision functions must be smooth under perturbation and cannot encode a large number of independent decision axes.

16.3 Influence Lower Bounds for Worst-Case NP Languages

We now show that such smoothness is impossible for worst-case NP decision.

Theorem 15 (Geometric Fragility of Worst-Case NP Decision)**.** *There exists an NP language L such that for every polynomial-time decider D for L, the induced decision function f_D satisfies*

$$\mathrm{Inf}(f_D) = \Omega(n),$$

and hence is noise sensitive under any constant noise rate $\varepsilon > 0$.

Final Criterion. The preceding analysis admits a concise formulation. Let $f_n : \{0,1\}^n \to \{0,1\}$ denote the Boolean decision function induced by a purported polynomial-time decider for an NP language, and let $N(n)$ denote the number of strongly independent verification constraints associated with instances of length n. Robustness under bounded perturbation is equivalent to noise stability, and hence implies bounded total influence, $\mathrm{Inf}(f_n) = O(1)$, which in turn bounds the mutual information between any internal computational state and the witness by a constant. Consequently, the total semantic information capacity of any robust polynomial-size computation is at most polynomial in n. For NP languages whose verification enforces $N(n) = \omega(\mathrm{poly}(n))$ independent constraints, this yields an incompatibility: robust decision would require representing super-polynomially many independent semantic degrees of freedom within a computation whose total information capacity is polynomially bounded. Equivalently, admissible efficient computation must satisfy

$$\limsup_{n\to\infty} \left(\mathrm{Stab}_\varepsilon(f_n) \cdot \frac{\log \mathcal{I}(f_n)}{N(n)} \right) > 0,$$

where $\mathrm{Stab}_\varepsilon(f_n)$ denotes noise stability under ε-perturbations and $\mathcal{I}(f_n)$ denotes the semantic information capacity of the computation. For worst-case NP languages, this condition fails. Thus, unless robustness is abandoned as a minimal requirement for meaningful efficient computation, polynomial-time decision cannot coincide with polynomial-time verification. This yields the separation $P \neq NP$.